Aditya Saraswat

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Education

Manipal University Jaipur — Expected 2028
B.Tech in Computer Science Engineering, Data Science Specialization — *Jaipur, India*

B.S. International School | *Class XII* — May 2024

Delhi Public School Indirapuram | *Class X* — Mar 2022

Experience

Backend Developer Intern — May 2026 – Jul 2026
Centre for Railway Information Systems (CRIS), Ministry of Railways — *New Delhi, India*

- Developed REST APIs and service-layer logic using Java, Spring Boot, and PostgreSQL for reservation chart and seat availability workflows.
- Implemented route-segment based seat filtering to validate station-to-station journeys and recommend accurate vacant seats.

Data Analytics Intern — Jun 2025 – Jul 2025
Centre for Railway Information Systems (CRIS), Ministry of Railways — *New Delhi, India*

- Analyzed large-scale Indian Railways freight data, built geospatial visualizations using Python, GeoPandas, and Plotly, and created Tableau dashboards for operational insights.

Projects

Finora AI – Cross-Platform Personal Finance Platform — In Progress
React, React Native, Spring Boot, Spring Cloud Gateway, PostgreSQL, Python, FastAPI, Docker, AWS

- Developing a cross-platform expense management platform for web and mobile, backed by Spring Boot microservices and REST APIs.
- Implementing JWT authentication, resource-level authorization, transaction management, and budget tracking across distributed services.
- Integrating a Python-based analytics service with ML-driven spending insights, forecasting, and LLM-powered natural-language financial queries.
- Automating containerized backend deployments and cloud infrastructure provisioning using AWS CDK, ECS, ECR, and GitHub Actions.

CredSight – Explainable Credit Risk Assessment — In Progress
Python, XGBoost, Scikit-learn, SHAP, FastAPI

- Developing an end-to-end credit default prediction pipeline with data validation, preprocessing, class-imbalance handling, and Logistic Regression and XGBoost model comparison.
- Optimizing XGBoost hyperparameters using randomized search and stratified cross-validation; implementing decision-threshold tuning and probability calibration for risk classification.
- Applying SHAP for global feature importance and individual prediction explanations; building a FastAPI REST endpoint and interactive web interface for real-time risk probability estimates.

Journal App — May 2026 – Present
Java, Spring Boot, MongoDB, Spring Security | *GitHub*

- Built a backend-focused journal management app with user management APIs and journal CRUD operations backed by MongoDB.
- Designed a clean layered architecture (controller, service, repository, model); adding Spring Security and JWT authentication.

Technical Skills

Languages: Java, Python, C, C++, JavaScript, SQL, HTML5, CSS3
Backend: Spring Boot, Spring Security, Spring Cloud Gateway, REST APIs, JWT Authentication, API Design, FastAPI
Frontend: React.js, React Native, Vite, Tailwind CSS, Responsive UI Design
Databases: MongoDB, PostgreSQL, MySQL, Database Design, Query Optimization
Machine Learning & Data: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost, SHAP, PyTorch
Cloud & DevOps: Docker, AWS (CDK, ECS, ECR), GitHub Actions, Vercel, Appwrite
Core CS: Data Structures and Algorithms, OOPS, DBMS, Operating Systems, Computer Networks
Tools: Git, GitHub, Postman, IntelliJ IDEA, VS Code, Tableau, GeoPandas, Plotly

Achievements and Certifications

- Solved 240+ LeetCode problems with Java-based solutions focused on DSA.
- ACM Hacks 9.0 Finalist – ranked among the top teams out of 60 participating teams.
- CCNA (3 courses): Introduction to Networks; Switching, Routing and Wireless Essentials; Enterprise Networking, Security and Automation – Cisco Networking Academy.
- Introduction to Modern AI; Python Essentials 1 – Cisco Networking Academy.
- Python for Data Science, AI and Development – IBM / Coursera; Python for Data Analysis: Pandas and NumPy – Coursera.
- Introduction to Front-End Development – Meta / Coursera; Programming in Java (Elite) – NPTEL.